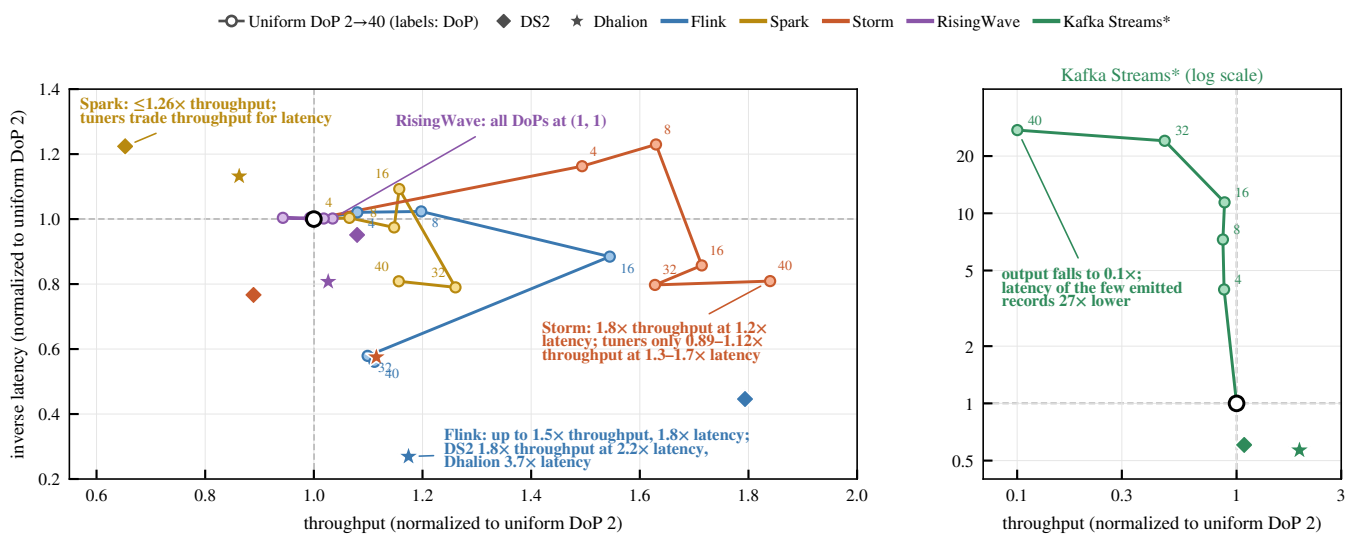
Uniform DoP 2→40 (labels: DoP)
DS2
Dhalion
Flink
Spark
Storm
RisingWave
Kafka Streams*
Spark: ≤1.26× throughput; tuners trade throughput for latency
RisingWave: all DoPs at (1, 1)
Storm: 1.8× throughput at 1.2× latency; tuners only 0.89–1.12× throughput at 1.3–1.7× latency
Flink: up to 1.5× throughput, 1.8× latency; DS2 1.8× throughput at 2.2× latency, Dhalion 3.7× latency
inverse latency (normalized to uniform DoP 2)
throughput (normalized to uniform DoP 2)
Kafka Streams* (log scale)
output falls to 0.1×; latency of the few emitted records 27× lower
throughput (normalized to uniform DoP 2)